

Definicire®

Good hair day



People make our name



The major role of hair lipids



Hair is naturally beautiful.

The best protection of the hair is that provided by nature. Sebum, a perfectly chemically designed natural hair conditioner, and the epicuticle, a lipid layer that includes 18-Methyl Eicosanoic Acid (18-MEA) provide lubricity to the hair and constitutes the first line of defense against environmental aggressions.

Frequent use of shampoo, heat, mechanical and chemical aggressions remove the protective lipidic film. Hair becomes consequently difficult to comb, rough to the touch, dull, but also prone to tangling and frizzing.

Silicones and quaternary compounds are the main cosmetic ingredients used as hair conditioners. They are both efficient, in different ways, but have limitations regarding biodegradability, build-up effect and image. In addition, they are not natural and are unlike the complex hair lipid layer.

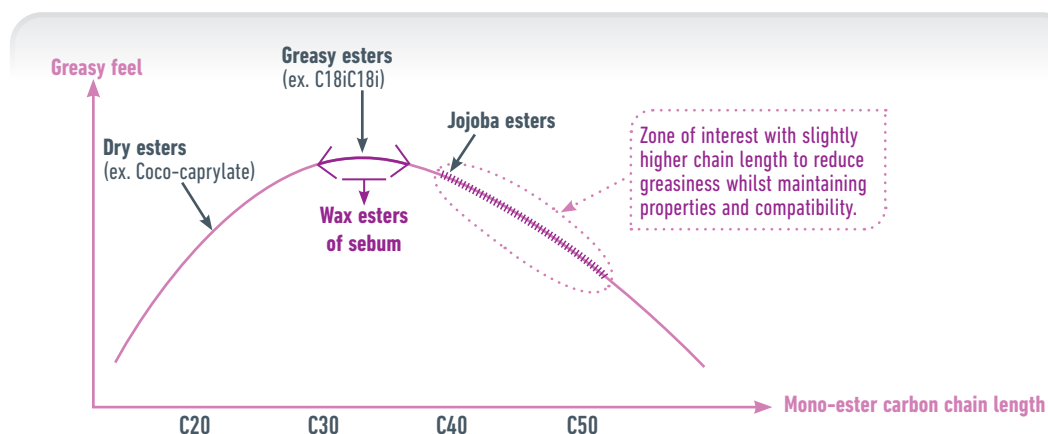
Gattefossé has placed its expertise in lipid chemistry at the service of haircare.

100% natural origin
COSMOS approved
Readily biodegradable (95%)*
Alternative to silicones & quaternary compounds

A biomimetic lipidic composition

The objective is to obtain a lipidic composition, inspired by natural hair lipids, that will not weigh down the hair or leave an unpleasant greasy sensation. This composition must rival the performance of quats and silicones and be suitable for different hair types.

The greasy feel of a lipidic ingredient is a function of the mono-ester carbon chain length as shown on the graph below. It allows the determination of the ideal chemical composition to combine performance with good sensory attributes.



Definicire® is a unique ingredient obtained by interesterification of jojoba wax and sunflower seed wax in presence of polyglycerol. Its composition has been optimized to recreate the performance of sebum and 18-MEA and contains:

JOJOBA ESTERS (C38-C44)

Close in chain length to sebum wax esters (C26-C42) but less oily and present good affinity with 18-MEA

MIXED ESTERS (C42-C64)

Synthesized via transesterification of jojoba and sunflower waxes, ensure hydrophobicity with low greasiness compared to sebum triglycerides and squalene

POLYGLYCEROL-3 ESTERS

Bring optimal spreadability similar to that provided naturally by sebum monoglycerides due to their amphiphilic behavior

* according to OECD 301F method

Perfect hair discipline in all climates

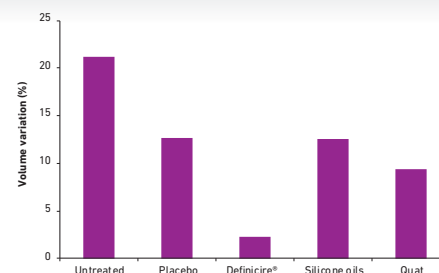
Healthy hair is hydrophobic. Definicire® reinforces the hydrophobicity of the hair. It acts as a lubricant decreasing friction between hair fibers and improves hair quality in both dry and humid conditions.

The benefits of Definicire®, used at 2% in a simple emulsion, have been evaluated in comparison with silicone oils and a quaternary ammonium used at the same concentration.



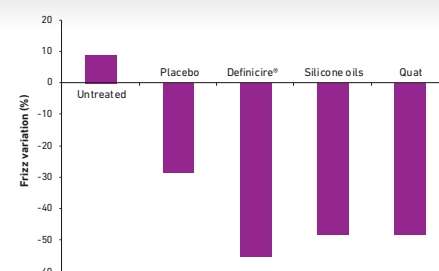
Once completely dry, the formulation containing Definicire® creates the most hydrophobic film. Moreover, due to optimal affinity and compatibility with hair lipids, Definicire® leaves a very smooth and homogeneous film on the hair surface.

Hair volume in humid conditions ⁽¹⁾



Definicire® provides a humidity barrier and helps to maintain the hairstyle.

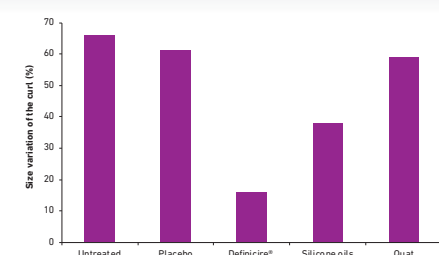
Frizz and fly-away hair ⁽¹⁾



Definicire® prevents the appearance of frizz and fly-aways.

Curl definition ⁽²⁾

When exposed to high temperatures and humidity levels, curls tend to fall flat. Size variation of the curl is a good indicator of its shape and definition in drastic climate conditions.



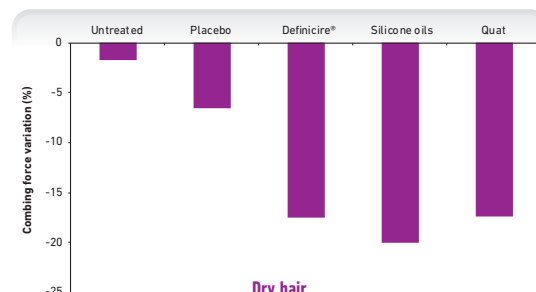
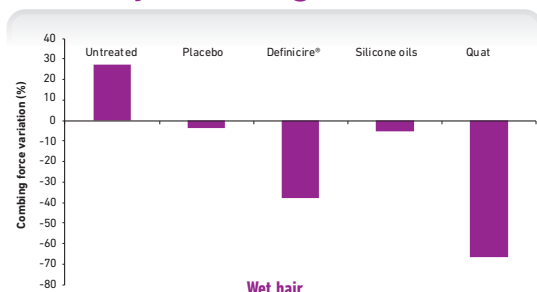
Definicire® helps retain curl shape and definition even in humid weather.

⁽¹⁾ Test performed on regular curly hair type using image analysis with the RUMBA device (Bossa Nova Technologies)

⁽²⁾ Assessed on straight hair type placed at 30°C and 60% relative humidity for 4 hours

True sensory benefits

Wet and dry combing ⁽³⁾

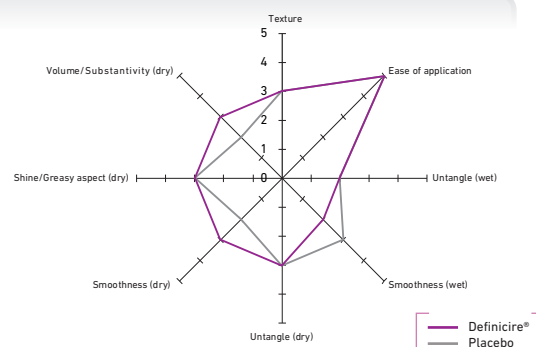


Definicire® makes hair combing significantly easier in both wet and dry conditions.

Expert sensorial evaluation ⁽⁴⁾

A trained technician has evaluated several attributes on hair tresses after application of a cream containing 2% of Definicare and a placebo formula.

A subjective scale was used to determine the difference between the applied products.



Definicire® enhances the smooth feel of wet and dry hair without imparting any greasiness or dull appearance. It has no untangling effect but it slightly increases hair substantivity. Moreover, the texture of the formulation and its application onto the hair are not modified by the addition of Definicare®.

Consumer test

Definicire® has been formulated at 2% in a leave-on anti-frizz serum and a rinse-off mask conditioner. Two groups of 10 volunteers, with different hair types, have evaluated the products over a period of one week. They were asked to give their opinion after the first use of the formula and again at the end of the test.

Anti-frizz serum ⁽⁵⁾		Mask conditioner ⁽⁶⁾	
My curls and waves feel unheavy	90%	My hair doesn't feel sticky	92%
My hair is soft and silky	80%	My hair shows good volume	83%
My hair is shiny	80%	My hair doesn't feel dry	100%
My hair doesn't feel sticky	100%	My hair is not greasy	82%
My hair doesn't feel dry	80%	Frizz and fly-aways remained controlled all day long	100%
My hair is not greasy	90%	My hair feels supple	92%
My hair is protected from static electricity	89%	Hair combing is easier	83%
My hair volume remained the same for the whole day	90%	My hairstyle is protected from static electricity	100%
The product is easy to use and pleasant during application	100%	Hair discipline was maintained during the day	92%
The texture is not sticky	100%	The texture is not sticky and not greasy	100%

Definicire® is adapted to all hair types and it offers key benefits to leave-on and rinse-off haircare products.

⁽³⁾ Evaluation performed on regular curly hair type using TA.XT Plus texture analyzer (Stable Micro Systems)

⁽⁴⁾ Assessed on straight Caucasian hair type

⁽⁵⁾ Frizz Control (2003-1.12)

⁽⁶⁾ Supreme Beautifying Hair Mask (2002-4.03)



Definicire[®]

is an active texturizing agent that recreates the hair protective lipid layer. 100% natural origin, readily biodegradable and based on functionalized jojoba and sunflower seed waxes, it ensures perfect hair discipline in all climates, and a comfortable smooth feel.

Jojoba wax, an arid-zone renewable resource, is the only liquid plant wax. It is extracted by cold pressing from the seeds of the shrub *Simmondsia chinensis*. Its yellow color is due to the presence of natural antioxidants that confer an excellent oxidative stability.

Sunflower seed wax is a pale colored hard wax present on seed shells. It is derived from the production of the edible oil. This vegetable wax, which is a good viscosity builder with outstanding oil binding capacity, shows excellent thermal stability.

Claims

- Increases hair hydrophobicity
- Ensures hair discipline: frizz and volume control, curl definition even under high humidity
- Inhibits the effect of static electricity
- Facilitates combing
- Enhances hair smoothness
- Protects from pollution

Applications

- Rinse-off conditioners and masks
- Pre-style treatments
- After-style: anti-frizz day creams, protective hair sprays and lotions
- Men's lines: beards, styling gums and waxes

Use level

2%

INCI name: Jojoba Esters (and) Helianthus Annuus (Sunflower) Seed Wax (and) Polyglycerin-3

Worldwide approval for cosmetic applications

Listed on IECIC (Inventory of Existing Cosmetic Ingredients in China) 2015

COSMOS approved

Readily biodegradable

Patent pending

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